

d-CCSSBU-L Chart Code and Its Output for Equipment A and B

1 Read Excel Data

Each Excel Data contains three spreadsheet: DAR, CAR, and Mic. Each spreadsheet should have variables Event, Location, Result, USL, LOD, LOQ (For Mic, it does not have LOD and LOQ).

2 d-CCSSBU-L Function

The function `d_ccssbu_l()` uses Excel data as above to generate D-CCSSBU-L chart.

3 Selection of d for d-CCSSBU-L Function

- `d_ccssbu_l()` constructs the d-CCSSBU-L chart for a specified value of d .
- `d_ccssbu_l_d_selection_arl1()` applies `d_ccssbu_l()` across all candidate values $d = 1, \dots, q$ using Excel input data and summarizes OOC counts and ARL_1 performance for DAR, CAR, and Mic residue types.
- `make_d_ccssbu_l_arl1_wide_tbl()` converts the output of `d_ccssbu_l_d_selection_arl1()` into a wide-format summary table.
- `render_d_ccssbu_l_arl1_tbl()` renders the output table from `make_d_ccssbu_l_arl1_wide_tbl()` as a responsive tibble that automatically adjusts to page width.
- `render_d_ccssbu_l_recommendation_tbl()` generates the recommended d selection table for each residue type (DAR, CAR, and Mic).

4 Application of the functions to Equipment A and B.

4.1 d-CCSSBU-L Selection of d and ARL1 Results for Equipment A

Applying the `d_ccssbu_l_d_selection_arl1()` function in [1] to Equipment A data produces a list of candidate values of d for each residue type. For a given residue type (e.g., DAR), the output contains q rows, where q is the number of ordered clusters identified.

For each candidate value of d , the first d clusters are selected to form the calibration dataset containing n_{Cal} cleaning events. The calibration dataset is then used in the stratified smoothed bootstrap procedure to estimate the chart UCL.

For the resulting UCL, the numbers of OOC events among ND events (OOC_ND), Stage 3A events (OOC_3A), and Stage 3B events (OOC_3B) are reported. An $\text{OOC_ND} > 0$ indicates that at least one ND event is identified as OOC. Because ND events represent the best-quality cleaning events, such a result suggests that the calibration dataset may not be representative of the IC process. Similarly, $\text{OOC_3A} > 0$ may indicate that the selected calibration dataset does not adequately represent an IC Stage 3A process.

For comparison with conventional control charts, the sampling-location size associated with each cleaning event is recorded, and the median value is used as the benchmark subgroup size. This benchmark subgroup size is used to construct the conventional $\bar{X}$ chart and S chart and to calculate the corresponding benchmark $\bar{X}$ -chart ARL_1 values reported in the table.

Finally, the d-CCSSBU-L chart ARL_1 values are calculated for a range of shift magnitudes. For the d-CCSSBU-L chart, the shift size is defined as $\delta_k = k \times \text{UCL}/3$, where k denotes the shift multiplier. The benchmark $\bar{X}$ -chart ARL_1 values are provided for comparison with the proposed chart.

Table 1: d-CCSSBU-L ARL1 simulation results for Equipment A. For each d , the d-CCSSBU-L chart was fitted using the location-specific calibration algorithm. ARL1 was estimated by shifting the bootstrap location statistics using delta $k(d)$ equal to k times $\text{UCLboot}(d)$ divided by 3.

Residue	d	q	case_label	n_{Cal}	OOC_ND	OOC_3A	OOC_3B	benchmark_n	$\bar{X}_{\text{bar_chart_ARL1_k_0p5}}$	UCL	ARL1_k_0p5	ARL1_k_1	ARL1_k_1p5	ARL1_k_2	ARL1_k_2p5	ARL1_k_3
CAR	1	1	Case 1	10	0	0	0	3	60.8923	4.1222	36.5631	36.5631	36.5631	35.9712	1.0000	1.0000
DAR	1	3	Case 2	3	0	0	0	6	26.3658	3.9132	6.8097	3.7064	2.2026	1.4708	1.0641	1.0000
DAR	2	3	Case 2	6	0	0	0	6	26.3658	4.5849	21.5285	7.2046	3.4764	2.0640	1.3702	1.0000
DAR	3	3	Case 2	10	0	0	0	6	26.3658	4.0912	50.7614	28.9855	9.1996	5.8428	2.5445	1.0002
Mic	1	1	Case 1	10	0	0	0	2	91.5158	2.0736	9.9751	9.9751	9.9751	9.9751	9.9751	1.2997

The recommendation algorithm recommended CAR $d=1$, DAR $d = 2$ by Case 1 in Step 8 although it totally has 3 clusters, and Mic $d = 1$.

Table 2: Recommended d values for the d-CCSSBU-L chart for Equipment A.

Residue	recommended_d	q	$\bar{X}_{\text{bar_chart_ARL1_k_0p5}}$	$D1_values$	$D2_values$	$D3_values$	$D4_values$	selected_case	selection_rule	D_star_values	selection_path	recommendation_notes
CAR	1	1	60.8923	1	1	1		Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	
DAR	2	3	26.3658	1, 2, 3	1, 2, 3	1, 2	3	Case 1	max(D1 intersect D2 intersect D3)	1, 2	Case 1: max(D1 intersect D2 intersect D3)	
Mic	1	1	91.5158	1	1	1		Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	

4.2 d-CCSSBU-L Selection of d and ARL1 Results for Equipment B

The `d_ccssbu_l_d_selection_arl1()` function was applied to the Equipment B data using the same procedure described for Equipment A.

For CAR, the candidate solution with $d = 1$ produced OOC ND events ($\text{OOC_ND} = 3$). Although all OOC observations were ND, the ordered-cluster algorithm identified two distinct groups of ND events, corresponding to relatively smaller and larger ND values. When only the first cluster was selected for calibration, the calibration dataset became dominated by the smaller ND values, resulting in an excessively small UCL. Consequently, several events containing larger ND values were identified as OOC.

This result illustrates that a calibration dataset should be representative of the IC process, even when all observations are ND. Therefore, the candidate solution with $d = 1$ was not considered suitable or the calibration dataset is not representative.

Table 3: d-CCSSBU-L ARL1 simulation results for Equipment B. For each d , the d-CCSSBU-L chart was fitted using the location-specific calibration algorithm. ARL1 was estimated by shifting the bootstrap location statistics using delta $k(d)$ equal to k times $\text{UCLboot}(d)$ divided by 3.

Residue	d	q	case_label	nCal	OOC_ND	OOC_3A	OOC_3B	benchmark_n	Xbar_chart_ARL1_k_0p5	UCL	ARL1_k_0p5	ARL1_k_1	ARL1_k_1p5	ARL1_k_2	ARL1_k_2p5	ARL1_k_3
CAR	1	2	Case 4	10	3	3	6	3	60.8923	2.1310	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
CAR	2	2	Case 4	20	0	0	0	3	60.8923	30.9387	500.0000	9.9701	6.3837	4.6083	3.5549	1.0000
DAR	1	26	Case 4	10	0	0	0	5	33.4221	19.5308	1.0088	1.0088	1.0088	1.0088	1.0088	1.0000
DAR	2	26	Case 5	13	0	0	0	5	33.4221	19.5454	1.2732	1.0051	1.0035	1.0032	1.0029	1.0000
DAR	3	26	Case 5	23	0	0	0	5	33.4221	23.4040	17.0648	3.4626	2.0206	1.3437	1.0264	1.0008
DAR	4	26	Case 5	29	0	0	0	5	33.4221	20.7417	6.8399	3.9131	2.9291	1.9916	1.0162	1.0000
DAR	5	26	Case 5	39	0	0	0	5	33.4221	20.7252	19.0476	7.7942	4.6992	2.7941	1.1529	1.0004
DAR	6	26	Case 5	49	0	0	0	5	33.4221	20.4053	31.4961	12.5156	7.5472	4.0875	1.3605	1.0014
DAR	7	26	Case 5	59	0	0	0	5	33.4221	19.9548	44.0529	16.5289	11.4548	6.8236	1.7751	1.0003
DAR	8	26	Case 5	69	0	0	0	5	33.4221	19.6397	64.7249	22.3714	17.2117	10.2564	2.4393	1.0001
DAR	9	26	Case 5	79	0	0	0	5	33.4221	19.5022	90.4977	30.3030	24.8756	14.8478	3.2321	1.0000
DAR	10	26	Case 5	89	12	9	0	5	33.4221	19.3354	133.3333	39.1389	34.5423	19.5122	4.2256	1.0001
DAR	11	26	Case 5	99	12	9	0	5	33.4221	19.2718	198.0198	51.9481	46.8384	27.1739	5.3262	1.0001
DAR	12	26	Case 5	109	12	9	0	5	33.4221	19.1111	312.5000	83.6820	76.3359	38.4615	8.2781	1.0009
DAR	13	26	Case 5	117	12	9	0	5	33.4221	18.9497	434.7826	97.5610	91.3242	42.2833	9.5283	1.0014
DAR	14	26	Case 5	124	12	9	0	5	33.4221	18.9689	500.0000	124.2236	118.3432	51.8135	10.8342	1.0010
DAR	15	26	Case 5	126	12	9	0	5	33.4221	19.0971	434.7826	119.0476	109.2896	60.9756	11.7716	1.0012
DAR	16	26	Case 5	136	12	9	0	5	33.4221	18.6654	666.6667	136.9863	135.1351	59.1716	13.7741	1.0011
DAR	17	26	Case 5	140	12	9	0	5	33.4221	19.1061	500.0000	144.9275	139.8601	77.5194	15.3139	1.0008
DAR	18	26	Case 5	150	12	9	0	5	33.4221	18.4172	666.6667	188.6792	186.9159	66.2252	17.0358	1.0013
DAR	19	26	Case 5	156	12	9	0	5	33.4221	14.2788	206.1856	206.1856	186.9159	37.3832	9.2166	1.0019
DAR	20	26	Case 5	162	12	9	0	5	33.4221	14.1476	210.5263	208.3333	194.1748	41.6667	10.6838	1.0023
DAR	21	26	Case 5	165	12	9	0	5	33.4221	16.2715	307.6923	202.0202	190.4762	44.4444	15.5280	1.0030
DAR	22	26	Case 5	175	15	10	0	5	33.4221	13.7396	289.8551	285.7143	263.1579	50.2513	10.9769	1.0036
DAR	23	26	Case 5	185	15	10	0	5	33.4221	13.9869	289.8551	289.8551	277.7778	53.0504	12.5000	1.0038
DAR	24	26	Case 5	190	15	10	0	5	33.4221	13.9279	307.6923	307.6923	289.8551	61.1621	14.0647	1.0000
DAR	25	26	Case 5	199	15	10	0	5	33.4221	13.8748	317.4603	317.4603	307.6923	60.7903	14.7820	1.0000
DAR	26	26	Case 5	206	15	10	0	5	33.4221	13.6825	434.7826	425.5319	400.0000	76.9231	16.0256	1.0091
Mic	1	1	Case 4	10	0	0	0	3	60.8923	4.0806	28.6944	28.6944	28.6944	28.6944	28.6944	1.7109

The following table summarizes the recommended values of d for Equipment B. For CAR and Mic, only one admissible candidate remained after applying the selection rules, resulting in recommended values of $d = 2$ and $d = 1$, respectively. For DAR, multiple admissible candidates were identified, and the selection procedure recommended $d = 6$, corresponding to the smallest ARL_1 among the admissible candidates.

Table 4: Recommended d values for the d-CCSSBU-L chart for Equipment B.

Residue	recommended_d	q	Xbar_chart_ARL1_k_0p5	D1_values	D2_values	D3_values	D4_values	selected_case	selection_rule	D_star_values	selection_path	recommendation_notes
CAR	2	2	60.8923	2	2	1	2	Case 6	min(D1 intersect D2 intersect D4)	2	Case 6: min(D1 intersect D2 intersect D4)	
DAR	6	26	33.4221	1, 2, 3, 4, 5, 6, 7, 8, 9	1, 2, 3, 4, 5, 6, 7, 8, 9	1, 2, 3, 4, 5, 6	7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26	Case 1	max(D1 intersect D2 intersect D3)	1, 2, 3, 4, 5, 6	Case 1: max(D1 intersect D2 intersect D3)	
Mic	1	1	60.8923	1	1	1	1	Case 1	max(D1 intersect D2 intersect D3)	1	Case 1: max(D1 intersect D2 intersect D3)	

- 61 1. Yang CX. D-CCSSBU-l-or-s control chart for cleaning validation [Internet]. GitHub repository; 2026. Available from: <https://github.com/ChandlerXiandeYang/d-CCSSBU-L-or-S-Control-Chart>